

```
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFD000 pte 0x21FD4813 pa 0x87F52000 perm 0x13
va 0x3FFFFFFE000 pte 0x21FD00C7 pa 0x87F40000 perm 0xC7
va 0x3FFFFFFF000 pte 0x20001C4B pa 0x80007000 perm 0x4B
print_pgtbl: OK
vmprint_test starting
page table 0x0000000087f22000
..0x0000000000000000: pte 0x0000000021fc7801 pa 0x0000000087f1e000
.. ..0x0000000000000000: pte 0x0000000021fc7401 pa 0x0000000087f1d000
.. .. ..0x0000000000000000: pte 0x0000000021fc7c5b pa 0x0000000087f1f000
.. .. ..0x00000000000001000: pte 0x0000000021fc701b pa 0x0000000087f1c000
.. .. ..0x00000000000002000: pte 0x0000000021fc6c17 pa 0x0000000087f1b000
.. .. ..0x00000000000003000: pte 0x0000000021fc6807 pa 0x0000000087f1a000
.. .. ..0x00000000000004000: pte 0x0000000021fc64d7 pa 0x0000000087f19000
..0x0000003fc0000000: pte 0x0000000021fc8401 pa 0x0000000087f21000
.. ..0x0000003ffffe0000: pte 0x0000000021fc8001 pa 0x0000000087f20000
.. .. ..0x0000003fffffd000: pte 0x0000000021fd4813 pa 0x0000000087f52000
.. .. ..0x0000003fffffe000: pte 0x0000000021fd00c7 pa 0x0000000087f40000
.. .. ..0x0000003ffffff000: pte 0x0000000020001c4b pa 0x0000000080007000
vmprint_test: done
pgaccess_test starting
pgaccess_test: OK
ugetpid_test starting
ugetpid_test: OK
```

```
print_pgtbl starting
va 0x0 pte 0x21FC7C5B pa 0x87F1F000 perm 0x5B
va 0x1000 pte 0x21FC701B pa 0x87F1C000 perm 0x1B
va 0x2000 pte 0x21FC6C17 pa 0x87F1B000 perm 0x17
va 0x3000 pte 0x21FC6807 pa 0x87F1A000 perm 0x7
va 0x4000 pte 0x21FC64D7 pa 0x87F19000 perm 0xD7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFD000 pte 0x21FD4813 pa 0x87F52000 perm 0x13
va 0x3FFFFFFE000 pte 0x21FD00C7 pa 0x87F40000 perm 0xC7
va 0x3FFFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_pgtbl: OK
vmprint_test starting
vmprint_test: done
pgaccess_test starting
pgaccess_test: OK
ugetpid_test starting
ugetpid_test: OK
pgtbltest: ksuper_test failed: not enough of super ptes, pid=3
$ 31771 zain aleem
```

```
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFD000 pte 0x21FEC413 pa 0x87FB1000 perm 0x13
va 0x3FFFFFFE000 pte 0x21FE7CC7 pa 0x87F9F000 perm 0xC7
va 0x3FFFFFFF000 pte 0x20001C4B pa 0x80007000 perm 0x4B
print_pgtbl: OK
vmprint_test starting
page table 0x000000087f81000
..0x0000000000000000: pte 0x0000000021fdf401 pa 0x0000000087f7d000
.. ..0x0000000000000000: pte 0x0000000021fdf001 pa 0x0000000087f7c000
.. .. ..0x0000000000000000: pte 0x0000000021fdf85b pa 0x0000000087f7e000
.. .. ..0x00000000000001000: pte 0x0000000021fdec1b pa 0x0000000087f7b000
.. .. ..0x00000000000002000: pte 0x0000000021fde817 pa 0x0000000087f7a000
.. .. ..0x00000000000003000: pte 0x0000000021fde407 pa 0x0000000087f79000
.. .. ..0x00000000000004000: pte 0x0000000021fde0d7 pa 0x0000000087f78000
..0x0000003fc0000000: pte 0x0000000021fe0001 pa 0x0000000087f80000
.. ..0x0000003fffe00000: pte 0x0000000021fdfc01 pa 0x0000000087f7f000
.. .. ..0x0000003fffffd000: pte 0x0000000021fec413 pa 0x0000000087fb1000
.. .. ..0x0000003fffffe000: pte 0x0000000021fe7cc7 pa 0x0000000087f9f000
.. .. ..0x0000003ffffff000: pte 0x0000000020001c4b pa 0x0000000080007000
vmprint_test: done
pgaccess_test starting
pgaccess_test: OK
ugetpid_test starting
ugetpid_test: OK
ksuper test: OK
```

```
tbltest      2 23 46504
console      3 24 0
pgtbltest
print_pgtbl starting
0x0 pte 0x21FCD45B pa 0x87F35000 perm 0x5B
0x1000 pte 0x21FD201B pa 0x87F48000 perm 0x1B
0x2000 pte 0x21FD1417 pa 0x87F45000 perm 0x17
0x3000 pte 0x21FD1007 pa 0x87F44000 perm 0x7
0x4000 pte 0x21FD1CD7 pa 0x87F47000 perm 0xD7
0x5000 pte 0x0 pa 0x0 perm 0x0
0x6000 pte 0x0 pa 0x0 perm 0x0
0x7000 pte 0x0 pa 0x0 perm 0x0
0x8000 pte 0x0 pa 0x0 perm 0x0
0x9000 pte 0x0 pa 0x0 perm 0x0
0x3FFFFFF6000 pte 0x0 pa 0x0 perm 0x0
0x3FFFFFF7000 pte 0x0 pa 0x0 perm 0x0
0x3FFFFFF8000 pte 0x0 pa 0x0 perm 0x0
0x3FFFFFF9000 pte 0x0 pa 0x0 perm 0x0
0x3FFFFFFA000 pte 0x0 pa 0x0 perm 0x0
0x3FFFFFFB000 pte 0x0 pa 0x0 perm 0x0
0x3FFFFFFC000 pte 0x0 pa 0x0 perm 0x0
0x3FFFFFFD000 pte 0x0 pa 0x0 perm 0x0
0x3FFFFFFE000 pte 0x21FC94C7 pa 0x87F25000 perm 0xC7
0x3FFFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_pgtbl: OK
print_test starting
print_test: done
pgaccess_test starting
tbltest: pgaccess_test failed: incorrect access bits set, pid=4
zain aleem 31771
```